PINNS for Feynman Integrals - Literature Review and All Theory Notes

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Chapter 1

Algebraic Geometry, Topology and Other Stuff

1.1 Topology

1.2 Rings

1.3 Lie Groups and Lie Algebras

Chapter 2

Graph Theory for Feynman Integrals

With various Feynman Integral representations, we work with different polynomials created by the propagators and vertex factors of the graph. These representations might be useful in future research to move through loss landscapes better, for this reason its a good idea to bring a really cool tool: Graph theory. Instead of describing these messed up polynomials in momentum space, we can just write them as sums of spanning trees/forests of Feynman graphs which makes our life easier to understand.

2.1 Basics of Graphs

This section describes some basic terminologies related to graphs which also serves to better understand where Feynman Integrals come from. This treatment follows Chapter 2 of Weinzierl's review [3]. First, we need to get some definitions out of the way:

1. Unoriented graph: Graph comprised of edges and vertices. Edges can connect vertices.

2. Connected or Disconnected: If any vertices have no edge attached to them, you have a disconnected graph. Feynman Diagrams represent connected graphs.

3. Quiver/Oriented Graph: Each edge has a direction. This is equivalent to specifying a vertex as a source or a sink.

4. Valency: A valency of a vertex is how many edges are attached to it. Vertices with valencies 0, 1, 2 are useful to distinguish.

5. Vertex, Valency = 0: This is a disconnected vertex, not very relevant to our application.

6. Vertex, Valency = 1: Any edge attached to this is known as an external edge. In physics, we generally ignore drawing these verticies (think about external momenta in Feynman Diagrams)

7. Vertex, Valency = 2: This is called a dot. This is also not very relevant because from a particle physics standpoint, it does not encode any meaningful interaction between particles.

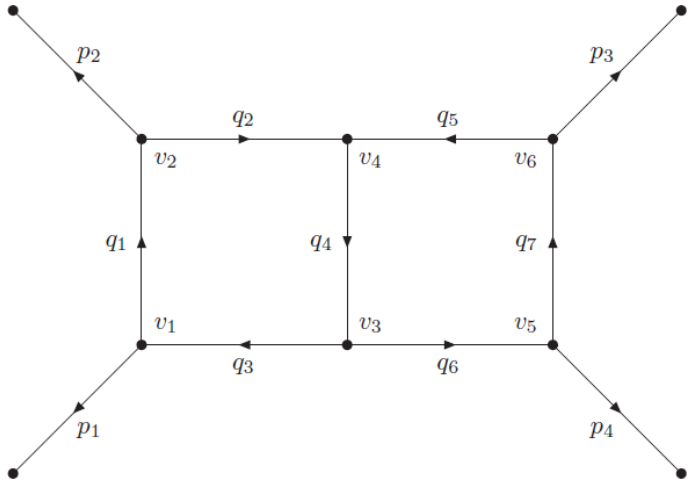

Figure 2.1: Example Feynman Graph.

We can see that a Feynman diagram is just a fancy graph with some additional information. Each edge has an orientation, an associated D-dimensional vector (p) and a number m. Important to note that the number m is not always mass because renormalization exists. The orientation of the edge depicts momentum flow and we don't draw external vertices.

Now, we can consider a graph G with n edges and r vertices. If the graph has k connected components, the loop number l of the graph is given by:

$$l = n - r + k \tag{2.1}$$

In mathematics lingo, this is called the first betti number of the graph of the cyclomatic number. We know that in physics, the loop number is equal to the number of independent momentum vectors not constrained by momentum conservation, so we basically have to integrate all of them through.
A connected graph of loop number 0 is called a **tree** and any graph of loop number 0 is called a **forest**. If the graph has k-connected components, with a loop number of 0, it is called a k-forest. Note that a tree is just a subset of a forest (wow so creative).
We now need to define some more notation :)

2.1.1 Notation

This section works only with connected graphs. Consider a graph G wiht n_{ext} external edges, n_{int} internal edges and loop number l. The graph will have vertices:

$$r = n_{ext} + n_{int} + 1 - l \rightarrow r_{int} = n_{int} + 1 - l \tag{2.2}$$

This is since we don't really deal with externals, which leaves us with r_{int} number of vertices with valency greater than 1. We give each edge an orientation (since its meant to represent Feynman integrals), which allows us to label edges such that $e_1, ..., e_{n_{int}}$ are

the internal edges and $e_{n_{int}+1}, ..., e_{n_{int}+n_{ext}}$ are the external edges. This is consistent with the type of notation we usually use for Feynman Integrals. Each edge has momentum flowing through it, denoted by q_j. For external momenta we define the usual redundant notation of external edge $e_{n_{int}+j}$ having momentum p_j giving us $p_j = q_{n_{int}+j}$.
Now moving towards vertices, consider a vertex v_a. $E^{source}(v_a)$ and $E^{sink}(v_a)$ denote guess what lmao, the set of edges which have a vertex v_a as source or sink. At each vertex we impose momentum conservation, which in graph theory lingo is written as:

$$\sum_{e_j \in E^{source}(v_a)} q_j = \sum_{e_j \in E^{sink}(v_a)} q_j \tag{2.3}$$

We denote the set of edges which have a vertex of valency 1 as source or sink as E^{in} or E^{out}. If we denote the independent loop momenta by k_i, we can express any momentum as a linear combination of the loop momenta and the external momenta:

$$q_j = \sum_{r=1}^{l} \lambda_{jr} k_r + \sum_{r=1}^{n_{ext}-1} \sigma_{jr} p_r \ ; \ \lambda_{jr}, \sigma_{jr} \in \{-1, 0, 1\} \tag{2.4}$$

As always, Feynman's $i0$ prescription is implied with the propagator terms wherever needed.

2.2 Spanning Trees and Spanning Forests

Let's start this analysis with a connected graph G with n edges and r vertices. Edges are labeled by $e_1, ..., e_n$ and the vertices by $v_1, ..., v_r$. Same notation as before. One discrepancy with how mathematicians and physicists approach graphs is the focus on internal edges, with the latter prefering to make the internal edges distinct. This is why we will bring in the internal graph G_{int} which is the sub-graph of G obtained by deleting the external vertices and edges from G. So basically cut the arms of the integral out.
We can start defining some things now:

Definition 2.2.1. A spanning tree for the graph G is a sub-graph T of G satisfying the following requirements:

1. T contains all the vertices of G
2. The loop number of T is 0.
3. T is connected

Basically, this spanning tree is just us squishing a feynman graph down until it got no loops. Just remove l edges corresponding to the loops. A general graph G can have multiple spanning trees.
Next is the spanning forest:

Definition 2.2.2. A spanning forest for the graph G, with respect to an edge set E is a sub-graph F of G satisfying:

1. F contains all the vertices of G

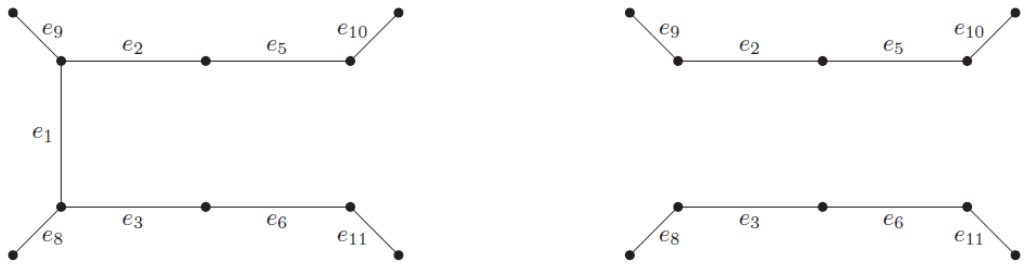

Figure 2.2: Spanning trees (right) and forests (left) for the example feynman graph shown previously.

2. The loop number of F is 0.
3. F contains all the edges $\{e_1, ..., e_n\} \backslash E$

This here is the superset of a spanning tree. First, notice how its not necessarily connected, it means you can now break down the original graph G however you want. However, for a connected feynman integral this runs the risk of ruining the physics, what if you get rid of one of the external momenta? This is why the with respect to an edge set has been inserted into this mix. In practice, the edge set is always the set of internal edges. This allows us to create k-forests of Feynman graphs G.
Basically, we get to decompose the Feynman Graph G into a bunch of building blocks using these spanning trees and forests. This can be seen in Figure 2.2.

We can denote the set of spanning forests of G by $\mathcal{T}$ and by $\mathcal{T}_k$ the set of spanning k-forests of G. We can write $\mathcal{T}$ as the union of all of these k-forests.

$$\mathcal{T} = \bigcup_{k=1}^{r_{int}} \mathcal{T}_k \tag{2.5}$$

This set decomposition makes the whole building blocks nature of these trees even more evident. Reminds me a lot of topology, wonder if there is a connection :)
We can write each element of $\mathcal{T}_k$ as:

$$(T_1, T_2, T_3, ..., T_k) \in \mathcal{T}_k \tag{2.6}$$

With each element being a connected component of the k-forest, which are necessarily trees. P_{T_i} denotes the set of external momenta associated with T_i.

2.2.1 Example of Trees and Forests

The Symanzik polynomials are used in Schwinger's representation of Feynman Integrals. These can be written easily in terms of the spanning trees and spanning 2-forests as:

$$\mathcal{F}(a) = \sum_{(T_1,T_2)\in\mathcal{T}_2} \left(\prod_{e_i \notin (T_1,T_2)} a_i \right) \left(\sum_{p_j \in P_{T_1}} \sum_{p_k \in P_{T_2}} \frac{p_j \cdot p_k}{\mu^2} \right) + \mathcal{U}(a) \sum_{i=1}^{n_{int}} a_i \frac{m_i^2}{\mu^2} \tag{2.7}$$

$$\mathcal{U}(a) = \sum_{T\in\mathcal{T}_1} \prod_{e_i \notin \mathcal{T}} a_i \tag{2.8}$$

Chapter 3

Differential Equations for Feynman Integrals

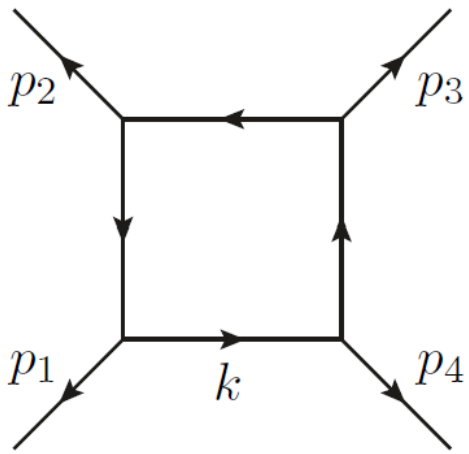

Figure 3.1: One-loop masless box integral, core example for this discussion.

3.1 Notation for Feynman Integrals

This discussion follows the one done in reference [1]. We will start with a background D-dimensional quantum field theory, in **momentum space**. Feynman integrals will then have integrals done over $\int d^d k$ with propagator factors like $1/[-(k+p)^2 + m^2 - i0]$. The $i0$ factor allows for a wick-rotation to be performed from Minkowski space to Euclidean space. In practice, this takes our path integral from looking like e^{iS} to e^{-S_E} in Euclidean space, which removes the oscillatory nature of the integral.
We will be dropping the $i0$ and assuming it to be present.

Now we can introduce the main example we are working with which is the box-integral which is seen in Figure 3.1. The Feynman diagram is translated into integral form using the Feynman rules which give us the mathematical form of the integral as:

$$I_{box} = \int \frac{d^d k}{i\pi^{D/2}} \frac{1}{k^2 (k+p_1)^2 (k+p_1+p_2)^2 (k-p^4)^2} \tag{3.1}$$

Each momentum is in D dimensions. We have four of them, and they all obey the on-shell condition $p_i^2 = 0$ and $\sum_{i=1}^{i-4} p_i = 0$.

However, in QFT, working with 4 external momentum vectors is not always the most efficient way to do things. Especially when you consider that these integrals are mostly evaluated for scattering problems, it is useful to switch variables to the Mandelstam variables given below:

$$s = (p_1 + p_2)^2 \ ; \ t = (p_2 + p_3)^2 \tag{3.2}$$

We can also see that Mandelstam variables are Poincare invariant, there is no need to prove it in a very mathematically rigorous way, we can just look at how they are defined. These are vectors squared, which means they are written as the actual of a dual vector on a vector, i.e they are just an inner product which by definition on a metric space is invariant.
For planar integrals, we can use dual coordinates which is just a change of variables.

$$I_{box} = \int \frac{d^D y}{i\pi^{D/2}} \frac{1}{(y-y_1)^2(y-y_2)^2(y-y_3)^2(y-y_4)^2} \tag{3.3}$$

Here, $y_{ij} = y_i - y_j$ and the kinematic invariants are written as $s = y_{13}^2$; $t = y_{24}^2$. This notation is very convenient because it allows us to write any one-loop Feynman integral (for example) in the form:

$$I_n = \int \frac{d^D y}{i\pi^{D/2}} \prod_{i=1}^{n} \frac{1}{[-(y-y_i)^2 + m_i^2]^{a_i}} \tag{3.4}$$

Note these indices a_i. These are the indices we are used to for classifying these integrals and what we really want to reduce.
The next step taken by the paper is to introduce the Feynman parameters, perform a Wick's rotation and run the integral over D-dimensions using the spherical coordinates. Its what we do when we first learn loop corrections, like shown in Sredniki's textbook [2]. We are going to use the generalized form of Feynman's formula:

$$\frac{1}{A_1^{\alpha_1} ... A_n^{\alpha_n}} = \frac{\Gamma(\sum_i \alpha_i)}{\prod_i \Gamma_{\alpha_i}} \frac{1}{(n-1)!} \int dF_n \frac{\prod_i x_i^{\alpha_i - 1}}{(\sum_i x_i A_i)^{\sum_i \alpha_i}} \tag{3.5}$$

This gives us a general representation of a one-loop integral:

$$I_n = \frac{\Gamma(a - D/2)}{\prod_i \Gamma(a_i)} \int_0^\infty \prod_{i=1}^{n} d\alpha_i \alpha_i^{a_i - 1} \frac{U^{a-D} \delta(\sum_i \alpha_i - 1)}{[V + U \sum_{i=1}^{n} \alpha_i m_i^2 - i0]^{a - D/2}} \tag{3.6}$$

Here, $V = \sum_{i<j} \alpha_i \alpha_j (-y_{ij}^2)$, $U = \sum_i \alpha_i$ and $a = \sum_i a_i$.

IMPORTANT NOTE: This notation isn't really that useful for our discussion but I am putting it out here for consistency with the review paper as they decompose the integral with this notation!

We can see how simply this formula works by looking at the box integral in $D = 6$ dimensions.

$$I_{box}^{D=6} = \int \prod_{i=1}^{4} d\alpha_i \frac{\delta(\sum \alpha_i - 1)}{\alpha_1 \alpha_3 (-s) + \alpha_2 \alpha_4 (-t)} \tag{3.7}$$

The above equation can be generalized even further for L-loops. This is shown below:

$$I_{n,L} = \frac{\Gamma(a - LD/2)}{\prod_i \Gamma(a_i)} \int_0^\infty \prod_{i=1}^n d\alpha_i \alpha_i^{a_i - 1} \frac{U^{a-(L+1)D/2} \delta(\sum_i \alpha_i - 1)}{[V + U \sum_{i=1}^n \alpha_i m_i^2]^{a - LD/2}} \tag{3.8}$$

With Feynman Integrals, its important to pay attention to the UV divergences (blowing up at high k) and IR divergences (region based singularities, etc.). These need to be regulated with dimensional regularization.
In all of the following discussion, the main focus is on evaluating Feynman Integrals for spacetime dimension $D = 4 - 2\epsilon$, so basically 4 spacetime dimensions.

3.2 IBP Reduction

Given a Feynman graph, we will now define a family of Feynman Integrals associated with it. The graph gives us the propagator structure, we will define the family of integrals as arbitrary powers of the propagators. With this notion of a family in mind we can look at the IBP relations.

We start by generalizing the box integral:

$$G_{a_1,a_2,a_3,a_4}(D; s, t) = \int \frac{d^D y}{i\pi^{D/2}} \prod_{i=1}^4 \frac{1}{[-(y - y_i)^2]^{a_i}} \tag{3.9}$$

Due to on-shell conditions: $y_{12}^2 = y_{23}^2 = y_{34}^2 = y_{41}^2 = 0$, which means the integral depends only on $s = y_{13}^2$ and $t = y_{24}^2$ alongside the dimension D. The set of G is the integral family. If we remove any of the propagators we get a triangle integral, if we give any propagators a negative power it goes into the numerator. The members of these integral families are not linearly independent in general and can be related to each other using a set of relations known as the IBP relations. These are essentially integration by parts applied to the integral. Dimensional regularization makes the total derivative vanish so we don't have to worry about the boundary term but we can work with these relation even if the boundary term is present.

Important Note: Its worth going through the mathematical proof of how dimensional regularization makes the total derivatives vanish. As it stands we can also argue that the whole IBP relation comes from the fact that total derivatives vanish and has nothing to do with integration by parts.

Mathematically, we can write the IBP relations as:

$$\int \frac{d^D y}{i\pi^{D/2}} \frac{\partial}{\partial y^\mu} \xi^\mu \prod_{i=1}^4 \frac{1}{[-(y - y_i)^2]^{a_i}} = 0 \tag{3.10}$$

Its worth taking some time to unpack this mathematical expression, which is something the review doesn't do properly. First, y_μ gives us the components of the y vector. ξ^μ is

an arbitrary vector which can be chosen to be $\xi^\mu = (y - y_1)^\mu$. Let's take the derivative acting on the propagator for y_1.

$$\frac{\partial}{\partial y^\mu}\frac{1}{[-(y-y_1)^2]^{a_2}} = \frac{-a_2}{[-(y-y_1)^2]^{a_2+1}}\frac{\partial}{\partial y^\mu}[-(y-y_1)^2] = \frac{-2a_2(y-y_1)_\mu}{[-(y-y_1)^2]^{a_2+1}}$$

When this is acted on by the vector $\xi = (y-y_1)$, we will just get $|y-y_1|^2$ as the numerator which cancels out one of the denominators. In this case its left the original but we will have the product rule acting on all four propagators to give different combinations, resulting in:

$$[(D - 2a_1 - a_2 - a_3 - a_4) - sa_3Y_3^+ + (-a_2Y_2^+ - a_3Y_3^+ - a_4Y_4^+)Y_1^-]G_{a_1,a_2,a_3,a_4} \tag{3.11}$$

Note that this is just what we get with $\xi = y - y_1$. We can generate all the cyclic combinations to get the full system of equations to solve. We can now spam these relations to reduce the value of a_i until it reaches zero. This is how we create the linearly independent basis. The linear independence is confirmed by the fact that they have already reached the last checkpoint for recursion, so they can't be related to any simpler integrals.

For higher loop integrals, these relations become really tough to solve so we use computational algorithms to solve this which are discussed later.

3.2.1 Example: Box Integral

We can IBP reduce the second last term to get the basis:

$$G_{2,1,1,1} = \frac{D-5}{s}G_{1,1,1,1} - \frac{4(D-5)(D-3)}{(D-6)st^2}G_{0,1,0,1} \tag{3.12}$$

$$G_{1,1,0,1} = \frac{2(D-3)}{(D-4)t}G_{0,1,0,1} \tag{3.13}$$

We can also think about this in Feynman Diagram terms as expressing a complicated loop diagram in terms of simple building block loop diagrams, with the loop number conserved in all.

3.3 Generating Differential Equations

We will denote the basis of an integral by $\vec{f}$. When we differentiate a basis $\vec{f}(s,t,\epsilon)$ with respect to s, t then we will end up getting a set of integrals that can be expressed back in terms of that basis. This is your differential equation, pretty simple.

$$\partial_s\vec{f}(s,t,\epsilon) = A_s(s,t,\epsilon)\vec{f}(s,t,\epsilon)\ ;\ \partial_t\vec{f}(s,t,\epsilon) = A_t(s,t,\epsilon)\vec{f}(s,t,\epsilon) \tag{3.14}$$

One caveat to keep in mind is that the differential operator ∂_s is not as straightforward as for momentum, the operator needs to be normalized and commute with the on-shell conditions. But if we express the Feynman integral in terms of the kinematic invariants, I don't see this being an issue.

3.3.1 Example: Box Integral

We previously IBP reduced the box integral a bit, and we can get 3 basis vectors:

$$f_1 = G_{0,1,0,1} \ ; \ f_2 = G_{1,0,1,0} \ ; \ f_3 = G_{1,1,1,1} \tag{3.15}$$

With these basis chosen we can get the differential equation spanned by the matrices:

$$A_s = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{-\epsilon}{s} & 0 \\ -\frac{2(1-2\epsilon)}{s(s+t)} & \frac{2(1-2\epsilon)}{s^2(s+t)} & -\frac{s+t+\epsilon t}{t(s+t)} \end{pmatrix} \tag{3.16}$$

$$A_t = \begin{pmatrix} -\frac{\epsilon}{t} & 0 & 0 \\ 0 & 0 & 0 \\ \frac{-2(1-2\epsilon)}{t^2(s+t)} & -\frac{2(1-2\epsilon)}{st(s+t)} & -\frac{s+\epsilon s+t}{t(s+t)} \end{pmatrix} \tag{3.17}$$

With this we have basically covered what we need to.

The paper then goes on to cover more topics about the differential equations such as finding canonical basis to factor out ϵ naturally and cutting your phase space to avoid singularities which are all very cool but as of right now, not directly relevant to the work we are trying to achieve.

3.4 Spurious and Unitarity Cuts

I want to cover this next ideally.

Chapter 4

Algorithms for Feynman Integral Solutions

The treatment of this section follows the work done in paper **CITE HERE, LAPORTA**. This is known as Laporta's algorithm. The paper discusses a variety of algorithms not just for integral reduction but also for solving master integrals using finite difference equations in one variable. This review will mostly just cover the reduction of the integral but I might extend it to solution of master integrals too.

4.1 Algorithm for Master Integral Hunting

We can first start by shifting our notation to that used by Laporta's paper.

4.1.1 Notation for Feynman Integrals

Consider a generic Feynman diagram wiht N_k loops, N_e external lines and N_d internal lines. The loop momenta are spanned by k_i and independent external momenta are p_i, spanned upto $N_p = N_e - 1$. Note that this is not the redundant, but useful notation used previously! The propagators have denominators $D_i = q_i + m_i^2$, where q_i can be either loop or external momentum or a linear combination of both, depending upon the propagator. The Feynman prescription for wick's rotations is implied.
In this notation, we write a generic Feynman integral as:

$$\int d^D k_1 d^D k_2 ... d^D k_{N_k} V_{\gamma\delta} \tag{4.1}$$

Here, the integrand is given by $V_{\gamma\delta}$:

$$V_{\gamma\delta} = \frac{\prod_{i=1}^{N_p} \prod_{j=1}^{N_k} (p_i.k_j)^{\delta_{ij1}} \prod_{i=1}^{N_k} \prod_{j=1}^{N_k} (k_i.k_j)^{\delta_{ij2}}}{\prod_{i=1}^{N_d} D_i^{\gamma_i}} \ ; \ \gamma_i \geq 0, \delta_{ijl} \geq 0 \tag{4.2}$$

IMPORTANT NOTE: This is a pretty big notation shift compared to the usual, we have an actual distinct notation for the numerator here. This is not a trivial shift. First off, these scalar products can't always be written as a linear combination or multiplication of propagators of the integral, say we have just a bare propagator, how will you write $k.p$ with just that? To fix that, in the other notations we have been using, we just make up some more propagators to just express these scalar products. However, in this notation

we write them explicitly as the numerator. This is a feature of the algorithm as it will be used later in sorting! This is why its a good idea to switch notations here.
The total number of these scalar products (N_{sp}) is given by:

$$N_{sp} = N_p N_k + \frac{N_k(N_k+1)}{2} \tag{4.3}$$

The final term is not N_k^2 because the symmetric products are the same, so we can just count them out and get the slightly skewed term. Think of the combinatorics like a matrix, so we will have the N_k diagonal square terms, then we will have the terms to one side of the matrix which will total $N_k^2/2 - N_k$ then we get the right term when we add it all up.

4.1.2 Notation for Identities

We are going to introduce something a bit new here, not standard IBP. Consider the integrand shown in Equation 5.2, we can write for each denominator D_j:

$$\frac{(p.k)_j}{D_j} = \frac{1}{C_j}\left(1 - \frac{D_j - C_j(p.k)_j}{D_j}\right) \ ; \ j = 1, ..., N_d \tag{4.4}$$

First, this is just an algebraic manipulation of a division. Its easy to see how this just works out to be the same shit. C_j can literally be any coefficient. Next is, why are we actually doing this? According to the paper this helps make each sent of irreducible scalar products correspond to a different combination of denominators. The correct choice of C_j also lets us strip the scalar product from the numerator entirely by properly cancelling it from D_j, thats why we have chosen the index j to be the same for both. This is why each scalar product $(p.k)_j$ must be chosen to be different for each application of this identity. When you replace each scalar product by denominator pair in the original integrand $V_{\gamma\delta}$, you can get after a bunch of multiplication, a sum of new terms in the general form:

$$V'_{ni\alpha\beta} = \frac{\prod_{j=1}^{N_{sp-n}} (p.k \ irred.)_j^{\beta_j}}{\prod_{j=1}^{n} D_{i_j}^{\alpha_j}} \tag{4.5}$$

With the above equation, we are only left with irreducible scalar products since we already removed all non-irreducible ones. The constants seem to be just factored out. The subscript is chosen to show dependencies. n stands for the number of denominators, i gives their particular combination $i = i_1, ..., i_n$ with the exponents $\alpha = \alpha_1, ..., \alpha_n$ and $\beta = \beta_1, ..., \beta_{N_{sp}-n}$.

This is an annoying process, but after everything is said and done, we are just left with nice building blocks of the integral with all numerators in their irreducible forms. All the reducible numerators are factored out and each set of scalar products has its own combination of denominators.

Now, we can finally write down the IBP identities in this notation, which act just on the building blocks:

$$\begin{gathered} \int d^D k ... d^D k_{N_k} \frac{\partial}{\partial (k_j)_\mu}((k_l)_\mu V'_{ni\alpha\beta}) \ ; j, l = 1, ..., N_k \\ \int d^D k ... d^D k_{N_k} \frac{\partial}{\partial (k_j)_\mu}((p_l)_\mu V'_{ni\alpha\beta}) \ ; \ j = 1, ..., N_k \ ; \ l = 1, ..., N_p \end{gathered} \tag{4.6}$$

The loop momenta vector will give N_k^2 identities since we can switch arround in any combination of loop momenta, the external momenta will give $N_k N_p$ identities through simple combinatorics. So the total number of IBP identities we have is $N_k(N_p + N_k)$ which is specified by the topology of the Feynman Integral.
The reducible scalar products that come from the action of the derivative on the building block term ALSO need to be made irreducible with the SAME process (this is such a mess lol) which results in a linear combination of integrals with all n denominators and with one denominator missing (this denominator is the one that can reduce the scalar product, which is now gone to make the product irreducible).

This has been a bit convoluted of a notation section, the main TLDR is: **We took a Feynman Integral with scalar products in the numerator and reduced it into a bunch of building blocks with only irreducible products using an algebraic identity that we then put into the IBP relations and reduced further to give IBP relations relating ONLY irreducible integrals**. Great, moving on.

Chapter 5

Proofs for Master Integrals

Chapter 6

Intersection Theory for Feynman Integrals

This is for the future!

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